Supplementary Material

This supplementary material provides detailed methodological information, extended data tables, and access to computational resources that support the conclusions of this scoping review.

S1. Detailed search strategy

Database	PubMed
Search date	July 16, 2025
Language	English
Publication period	01/2015 - 07/2025
Filters applied	Full articles available in English, published between 2015 and 2025.
Study type included	Primary studies with application of data mining techniques in pharmacovigilance databases (excluding reviews, editorials, letters, etc.).
Search results	161 articles
Search query	*(* *"Pharmacovigilance"[Majr]* *OR "Drug Monitoring/adverse effects"[Majr]* *OR "Drug-Related Side Effects and Adverse Reactions"[Majr]* *OR "Monitoring, Drug"[All Fields]* *OR "Therapeutic Drug Monitoring"[All Fields]* *OR "Drug Monitoring, Therapeutic"[All Fields]* *OR "Monitoring, Therapeutic Drug"[All Fields]* *OR "Drug Related Side Effects and Adverse Reactions"[All Fields]* *OR "Side Effects of Drugs"[All Fields]*

	OR "Drug-Related Side Effects and Adverse Reaction"[All Fields] *OR "Adverse Drug Reaction"[All Fields]* *OR "Adverse Drug Reactions"[All Fields]* *OR "Drug Reaction, Adverse"[All Fields]* *OR "Drug Reactions, Adverse"[All Fields]* *OR "Reactions, Adverse Drug"[All Fields]* *OR "Adverse Drug Event"[All Fields]* *OR "Adverse Drug Events"[All Fields]* *OR "Drug Event, Adverse"[All Fields]* *OR "Drug Events, Adverse"[All Fields]* *OR "Drug Side Effects"[All Fields]* *OR "Drug Side Effect"[All Fields]* *OR "Side Effect, Drug"[All Fields]* *OR "Side Effects, Drug"[All Fields]* *OR "Drug Toxicity"[All Fields]* *OR "Toxicity, Drug"[All Fields]* *OR "Drug Toxicities"[All Fields]* *OR "Toxicities, Drug"[All Fields]* *)* *AND* *(* *"Data Mining"[Majr]* *OR "Mining, Data"[All Fields]* *OR "Text Mining"[All Fields]* *OR "Mining, Text"[All Fields]* *)*

Database	Scopus
Search date	July 16, 2025
Language	English
Publication period	01/2015 - 07/2025
Filters applied	Full articles available in English, published between 2015 and 2025.
Study type included	Primary studies with application of data mining techniques in pharmacovigilance databases (exclusion of reviews, editorials, letters, etc.).
Search results	543 articles
Search query	((TITLE-ABS-KEY("pharmacovigilance") OR TITLE-ABS-KEY("drug monitoring") OR TITLE-ABS-KEY("therapeutic drug monitoring") OR TITLE-ABS-KEY("monitoring, drug") OR TITLE-ABS-KEY("drug monitoring, therapeutic") OR TITLE-ABS-KEY("monitoring, therapeutic drug") OR TITLE-ABS-KEY("drug related side effects and adverse reactions") OR TITLE-ABS-KEY("side effects of drugs") OR TITLE-ABS-KEY("drug-related side effects and adverse reaction") OR TITLE-ABS-KEY("adverse drug reaction") OR TITLE-ABS-KEY("adverse drug reactions") OR TITLE-ABS-KEY("drug reaction, adverse") OR TITLE-ABS-KEY("drug reactions, adverse") OR TITLE-ABS-KEY("reactions, adverse drug") OR TITLE-ABS-KEY("adverse drug event") OR TITLE-ABS-KEY("adverse drug events") OR TITLE-ABS-KEY("drug event, adverse") OR TITLE-ABS-KEY("drug events, adverse") OR TITLE-ABS-KEY("drug

	side effects") OR TITLE-ABS-KEY("drug side effect") OR TITLE-ABS-KEY("side effect, drug") OR TITLE-ABS-KEY("side effects, drug") OR TITLE-ABS-KEY("drug toxicity") OR TITLE-ABS-KEY("toxicity, drug") OR TITLE-ABS-KEY("drug toxicities") OR TITLE-ABS-KEY("toxicities, drug")) AND (TITLE-ABS-KEY("data mining") OR TITLE-ABS-KEY("text mining") OR TITLE-ABS-KEY("mining, data") OR TITLE-ABS-KEY("mining, text")))

Database	Web Of Science
Search date	July 16, 2025
Language	English
Publication period	01/2015 - 07/2025
Filters applied	Full articles available in English, published between 2015 and 2025.
Study type included	Primary studies with application of data mining techniques in pharmacovigilance databases (excluding reviews, editorials, letters, etc.).
Search results	339 articles
Search query	*((TS=("pharmacovigilance") OR TS=("drug monitoring") OR TS=("therapeutic drug monitoring") OR TS=("monitoring, drug") OR TS=("drug monitoring, therapeutic") OR TS=("monitoring, therapeutic drug") OR TS=("drug related side effects and adverse reactions") OR TS=("side effects of drugs") OR TS=("drug-related side effects and*

	adverse reaction") OR TS=("adverse drug reaction") OR TS=("adverse drug reactions") OR TS=("drug reaction, adverse") OR TS=("drug reactions, adverse") OR TS=("reactions, adverse drug") OR TS=("adverse drug event") OR TS=("adverse drug events") OR TS=("drug event, adverse") OR TS=("drug events, adverse") OR TS=("drug side effects") OR TS=("drug side effect") OR TS=("side effect, drug") OR TS=("side effects, drug") OR TS=("drug toxicity") OR TS=("toxicity, drug") OR TS=("drug toxicities") OR TS=("toxicities, drug")) AND (TS=("data mining") OR TS=("text mining") OR TS=("mining, data") OR TS=("mining, text")))

Database	Embase
Search date	July 16, 2025
Language	English
Publication period	01/2015 - 07/2025
Filters applied	Full articles available in English, published between 2015 and 2025.
Study type included	Primary studies with application of data mining techniques in pharmacovigilance databases (excluding reviews, editorials, letters, etc.).
Search results	425 articles
Search query	*('pharmacovigilance'/exp OR 'drug monitoring'/exp OR 'therapeutic drug monitoring':ti,ab OR 'monitoring, drug':ti,ab OR 'drug monitoring, therapeutic':ti,ab OR 'monitoring, therapeutic drug':ti,ab OR 'drug*

	related side effects and adverse reactions':ti,ab OR 'side effects of drugs':ti,ab OR 'drug-related side effects and adverse reaction':ti,ab OR 'adverse drug reaction':ti,ab OR 'adverse drug reactions':ti,ab OR 'drug reaction, adverse':ti,ab OR 'drug reactions, adverse':ti,ab OR 'reactions adverse drug':ti,ab OR 'adverse drug event':ti,ab OR 'adverse drug events':ti,ab OR 'drug event, adverse':ti,ab OR 'drug events, adverse':ti,ab OR 'drug side effects':ti,ab OR 'drug side effect':ti,ab OR 'side effect, drug':ti,ab OR 'side effects, drug':ti,ab OR 'drug toxicity':ti,ab OR 'toxicity drug':ti,ab OR 'drug toxicities':ti,ab OR 'toxicities drug':ti,ab) AND ('data mining'/exp OR 'mining, data':ti,ab OR 'text mining':ti,ab OR 'mining, text':ti,ab)

S2. Detailed characteristics of included studies (n = 162)

This table provides study-level information on data sources, analytical techniques, therapeutic focus, and data availability for all included articles. The references of the included studies are provided in the main reference list of the manuscript.

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
1	2024	FAERS	Information Component, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Chi-Square Test	J01AA08 (Minocycline), J01BA01(Chloramphenicol), J01MA14 (Moxifloxacin)	2004	2022	no	India	yes
2	2024	FAERS	Reporting Odds Ratio	Not used	Chi-Square Test	A10BA02 (Metformin), A10BK (SGLT-2 Inhibitors)	2013	2023	no	Türkiye	yes
3	2024	FAERS	Empirical Bayes Geometric Mean, Information Component, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	N05BA01 (Diazepam), N05BA04 (Oxazepam), N05BA06 (Lorazepam), N05BA08 (Bromazepam), N05BA12(Alprazolam), N05BB01 (Hydroxyzine), N05BC01 (Meprobamate), N05CD08 (Midazolam)	1990	2020	no	Pakistan	no
4	2024	VigiBase	Bayesian Confidence Propagation Neural Network, Information Component, Reporting Odds Ratio	Not used	Effect Size Calculation, Funnel Plot	D11AH05 (Dupilumab)	1978	2021	yes	Saudi Arabia	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
5	2020	FAERS	Information Component, Proportional Reporting Ratio	Not used	Adverse Outcome Pathways (AOPs) Analysis, Cross-Talk Analysis, Pathway Functional Enrichment Analysis	N05 (Psycholeptics), N06 (Psychoanaleptics)	1968	2018	yes	Greece	no
6	2019	VigiLyze	Information Component	Not used	Co-report Analysis (Concurrent Drugs), Geographic and Demographic Analysis	A10BG03 (Pioglitazone)	1967	2018	yes	India	no
7	2021	FAERS	Bayesian Confidence Propagation Neural Network, Empirical Bayes Geometric Mean, Information Component, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Chi-Square Test, Descriptive Statistics, Fisher's Exact Test, Mann-Whitney U Test	D11AH05 (Dupilumab), R03DX05 (Omalizumab)	2004	2021	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
8	2025	FAERS	Reporting Odds Ratio, Proportional Reporting Ratio, Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker	Not used	Time-to-Onset (TTO) / Time-to-Event Analysis, Serious Outcomes Rate	L01FF03 (Durvalumab)	2017	2024	yes	China	no
9	2020	FAERS	Bayesian Confidence Propagation Neural Network, Information Component, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Chi-Square Test, Descriptive Statistics, Fisher's Exact Test, Kruskal-Wallis Test, Mann-Whitney U Test	L01FF01 (Nivolumab), L01FF02 (Pembrolizumab), L01FF03 (Durvalumab), L01FF04 (Avelumab), L01FF05 (Atezolizumab), L01FX04 (Ipilimumab)	2004	2019	yes	China	no
10	2022	FAERS	Bayesian Confidence Propagation Neural Network, Empirical Bayes Geometric Mean, Information Component, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Chi-Square Test, Kruskal-Wallis Test	A10BK (SGLT-2 Inhibitors)	2004	2020	yes	China	no
11	2022	FAERS	Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Chi-Square Test	J01XX09 (Daptomycin)	2004	2021	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
12	2022	FAERS	Case/non-case analysis, Reporting Odds Ratio	Not used	Bonferroni Adjustment, Chi-Square Test, Propensity Score Matching (PSM)	L01FF01 (Nivolumab), L01FF02 (Pembrolizumab), L01FF03 (Durvalumab), L01FF04 (Avelumab), L01FF05 (Atezolizumab), L01FX04 (Ipilimumab)	2004	2020	yes	China	no
13	2024	FAERS	Empirical Bayes Geometric Mean, Information Component, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Chi-Square Test, Descriptive Statistics	L01XX73 (sotorasib), L01XX77 (adagrasib)	2021	2024	yes	China	no
14	2024	FAERS	Information Component, Reporting Odds Ratio	Not used	Austin Bradford-Hill Criteria, Logistic Regression, Receptor Occupancy Calculation / pKi	N05A (Antipsychotics)	2004	2023	yes	China	no
15	2020	KIDS-KD/ KAERS	Information Component, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Chi-Square Test	N03AX11 (topiramate)	2010	2017	yes	South Korea	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
16	2023	FAERS	Bayesian Confidence Propagation Neural Network, Information Component, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Chi-Square Test, False Discovery Rate (FDR), Time-to-Onset (TTO) / Time-to-Event Analysis, Weibull Shape Parameter (WSP) Test	L01CB01 (etoposide)	2004	2022	yes	China	no
17	2024	FAERS	Bayesian Confidence Propagation Neural Network, Empirical Bayes Geometric Mean, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Not used	L04AD01 (Ciclosporin)	2013	2022	yes	China	no
18	2018	FAERS	Proportional Reporting Ratio	Association Rule Mining (ARM) – Apriori	Confidence, Lift, Support	N02BE01 (paracetamol), N03AE01 (Clonazepam), N05AH04(quetiapine), N05BA12(Alprazolam), N05CF02(zolpidem), N06AB05 (paroxetine), N07BA03 (varenicline)	2004	2015	yes	United States of America	no
19	2022	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Hospitalization Rate, Survival Analysis / Mortality Rate, Time-to-Onset (TTO) /	A10BJ (Glucagon-like peptide-1 (GLP-1) analogues)	2004	2021	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
					Time-to-Event Analysis						
20	2024	FAERS	Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Time-to-Onset (TTO) / Time-to-Event Analysis	L04AC10(secukinumab)	2015	2021	yes	India	no
21	2019	FAERS	Empirical Bayes Geometric Mean, Proportional Reporting Ratio, Reporting Odds Ratio, Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker	Not used	Mann-Whitney U Test, Survival Analysis / Mortality Rate, Time-to-Onset (TTO) / Time-to-Event Analysis	L01FF01 (Nivolumab), L01FF02 (Pembrolizumab), L01FF03 (Durvalumab), L01FF04 (Avelumab), L01FF05 (Atezolizumab), L01FX04 (Ipilimumab)	2004	2018	yes	China	no
22	2024	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics, Serious Outcomes Rate, Time-to-Onset (TTO) / Time-to-Event Analysis	J05AB09(famciclovir)	2004	2023	yes	China	no
23	2025	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	L01EX19 (ripretinib)	2020	2024	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techiniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
24	2015	FAERS	Information Component, Reporting Odds Ratio	Not used	Event Sequence Symmetry Analysis (ESSA), Sequence Ratio (SR) / Adjusted Sequence Ratio	C10AA (HMG CoA reductase inhibitors)	2004	2012	yes	Japan	no
25	2016	FAERS	Reporting Odds Ratio	Not used	Geographic and Demographic Analysis	N06AA04 (clomipramine), N06AA09(amitriptyline), N06AX03 (mianserin), N06AX14 (tianeptine), N06AX22(agomelatine)	1968	2014	yes	Germany	no
26	2025	FAERS	Bayesian Confidence Propagation Neural Network, Empirical Bayes Geometric Mean, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	D11AH05 (Dupilumab)	2017	2023	yes	China	no
27	2023	FAERS	Bayesian Confidence Propagation Neural Network, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	J05AB01 (aciclovir), J05AB06 (ganciclovir), J05AB09(famciclovir), J05AB11 (valaciclovir)	2004	2022	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
28	2020	KIDS-KD/ KAERS	Information Component, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Causality Analysis (WHO-UMC), Logistic Regression, Time-to-Onset (TTO) / Time-to-Event Analysis	L04AB01 (etanercept), L04AB02(infliximab), L04AB04 (adalimumab)	2005	2016	yes	South Korea	no
29	2024	FAERS	Bayesian Confidence Propagation Neural Network, Empirical Bayes Geometric Mean, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	L02AE02 (leuprorelin)	2004	2023	yes	China	no
30	2017	FAERS	Information Component, Reporting Odds Ratio	Association Rule Mining (ARM) – Apriori	Time-to-Onset (TTO) / Time-to-Event Analysis, Weibull Shape Parameter (WSP) Test	C08 (calcium Channel Blockers), L04AA06 (mycophenolic acid), L04AD01 (Ciclosporin), L04AH01 (sirolimus), L04AH02(everolimus), N03AB02 (phenytoin), N03AF01 (carbamazepine), N03AG01 (valproic acid)	2004	2016	yes	Japan	no
31	2024	FAERS	Reporting Odds Ratio	Not used	Chi-Square Test	M05BX04(denosumab)	2010	2023	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
32	2023	KIDS-KD/ KAERS	Case-Only Odds Ratio (OR) Analysis, Likelihood Ratio Test	Tree-based Scan Statistic (TreeScan), Logistic Regression	McNemar Test, Wald Test	C02-3-7-8-9-10 (Antihypertensive Medicines)	2016	2020	no	Korea	no
33	2020	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Chi-Square Test, Dunn Index, Fisher's Exact Test, Kruskal-Wallis Test, Survival Analysis / Mortality Rate, Time-to-Onset (TTO) / Time-to-Event Analysis	L01FF01 (Nivolumab), L01FF02 (Pembrolizumab), L01FF03 (Durvalumab), L01FF04 (Avelumab), L01FF04(Cemiplimab), L01FF05 (Atezolizumab), L01FX04 (Ipilimumab)	2004	2019	yes	China	no
34	2024	FAERS, VigiAccess	Reporting Odds Ratio	Not used	Chi-Square Test	N06DX04 (Lecanemab), N06DX03 (Aducanumab)	Database inception	2024	yes	China	no
35	2025	FAERS	MHRA Method, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	D11AH08 (abrocitinib)	2022	2024	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techiniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
36	2024	FAERS	Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics, In Silico Computational Analysis / Molecular Docking	N01AB08(sevoflurane)	2004	2022	yes	India	no
37	2025	FAERS	Combination Risk Ratio Model (CRR), Shrinkage Measure Model	Not used	Cohen Kappa Coefficient, Chi-Square Test	N05A (Antipsychotics), J01 (Antibacterials for systemic use), L01 (Antineoplastic agents), C01B (Antiarrhythmics), N06A (Antidepressants), N02B (Analgesics)	2004	2023	yes	China	no
38	2025	FAERS	Bayesian Confidence Propagation Neural Network, Empirical Bayes Geometric Mean, Information Component, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Not used	N06AX29(brexanolone)	2019	2024	yes	China	no
39	2025	FAERS	Bayesian Confidence Propagation Neural Network, Reporting Odds Ratio	Not used	Descriptive Statistics	L01EX10(Midostaurin), L01EX13 (Gilteritinib)	2017	2023	yes	china	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
40	2024	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	N05AH05 (Asenapine)	2009	2023	yes	china	no
41	2024	FAERS	Information Component, Proportional Reporting Ratio, Reporting Odds Ratio, Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Empirical Bayes Geometric Mean	Not used	Chi-Square Test, Time-to-Onset (TTO) / Time-to-Event Analysis	L01FX05(brentuximab vedotin)	2012	2023	yes	china	no
42	2021	VigiBase	Bayesian Confidence Propagation Neural Network, Information Component, Case-Only Odds Ratio (OR) Analysis	Not used	Descriptive Statistics	D11AH05 (Dupilumab)	Database inception	2021	no	Israel	no
43	2017	KIDS-KD/ KAERS	Information Component, Proportional Reporting Ratio, Reporting Odds Ratio, Bayesian Confidence Propagation Neural Network	Not used	Not used	N06AB03 (fluoxetine)	1988	2015	no	Korea	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
44	2024	FAERS	Information Component, Proportional Reporting Ratio	Not used	Chi-Square Test, Descriptive Statistics	L01FA01(rituximab), L01FDO1(Trastuzumab), L01FF01 (Nivolumab), L01FF02 (Pembrolizumab), L01FF05 (Atezolizumab), L01FG01(Bevacizumab), L01FX04 (Ipilimumab), L01FX07(Daratumumab), M05BX04(denosumab), L01FE02 (Panitumumab)	Not informed	2023	yes	India	no
45	2023	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	L01EX07(Cabozantinib), L01EX08(Lenvatinib), L01XE05 (Sorafenib), L01XE21(Regorafenib)	2004	2021	yes	China	no
46	2025	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Time-to-Onset (TTO) / Time-to-Event Analysis	J05AX25 (Baloxavir Marboxil)	2018	2023	yes	China	no
47	2024	FAERS	Reporting Odds Ratio, Proportional Reporting Ratio	Not used	Chi-Square Test, Descriptive Statistics	R01AD03 (Triamcinolone Acetonide), R03BA01 (Beclometasone), R03BA02 (Budesonide), R03BA05 (Fluticasone), R03BA07 (Mometasone), R03BA08 (Ciclesonide)	2015	2023	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
48	2021	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Chi-Square Test, Kruskal-Wallis Test, Logistic Regression, Mann-Whitney U Test	L01CD01 (Paclitaxel), L01CD02 (Docetaxel), L01CD04 (Nab-paclitaxel)	2004	2019	yes	China	no
49	2023	FAERS	Empirical Bayes Geometric Mean, Relative Reporting Ratio	Network Analysis / Hierarchical Clustering Analysis	Jaccard Distance Calculation, RxNorm Normalization	N01AH07 (Meperidine), N02AA01 (Morphine), N02AA03 (Hydromorphone), N02AA05 (Oxymorphone), N02AA08 (Codeine), N02AB03 (Fentanyl), N02AB04 (Hydrocodone), N02AB05 (Oxycodone), N02AX02 (Tramadol), N02AX09 (Tapentadol), N07BC01 (Buprenorphine), N07BC02 (Methadone), R05DA03 (Dihydrocodeine)	2004	2020	yes	United States of America	no
50	2024	FAERS	Bayesian Confidence Propagation Neural Network, Empirical Bayes Geometric Mean, Information Component, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	C10AX16 (Bempedoic Acid)	2020	2023	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
51	2025	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	J02AC04(voriconazole)	2004	2024	yes	China	no
52	2025	FAERS	Bayesian Confidence Propagation Neural Network, Empirical Bayes Geometric Mean, Information Component, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Chi-Square Test, Descriptive Statistics	L01EX17 (Capmatinib)	2020	2023	yes	China	no
53	2024	FAERS	Reporting Odds Ratio	Not used	Chi-Square Test, Descriptive Statistics	C10AX12 (Alirocumab), C10AX13 (Evolocumab)	2015	2022	yes	China	no
54	2023	FAERS	MHRA Method, Reporting Odds Ratio, Bayesian Confidence Propagation Neural Network, Proportional Reporting Ratio	Not used	Descriptive Statistics, Chi-Square Test	N03AX22 (Perampanel)	2012	2022	yes	China	no
55	2024	FAERS	Bayesian Confidence Propagation Neural Network, Information Component, Multi-item Gamma Poisson Shrinker, Reporting Odds Ratio	Not used	Chi-Square Test, Descriptive Statistics, Time-to-Onset (TTO) /	A07EC02 (Mesalazine)	2018	2022	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
					Time-to-Event Analysis						
56	2024	FAERS	Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis, Volcano Plot	M05BX04(denosumab)	2010	2022	yes	China	no
57	2024	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio, Information Component	Not used	Descriptive Statistics, Chi-Square Test	N05CM18 (Dexmedetomidine)	2004	2023	yes	China	no
58	2025	JADER, FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Reporting Odds Ratio, Proportional Reporting Ratio	Not used	Descriptive Statistics, Kaplan-Meier method, Log-rank test, Time-to-Onset (TTO) / Time-to-Event Analysis, Weibull Shape Parameter (WSP) Test	N05AX13 (Paliperidone Palmitate)	2009	2024	no	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techiniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
59	2024	FAERS	Bayesian Confidence Propagation Neural Network, Proportional Reporting Ratio, Reporting Odds Ratio, Information Component	Not used	Descriptive Statistics	N05A (Antipsychotics), L04A (Immunosuppressants), N02B (Analgesics), H02A (Corticosteroids for systemic use), J01 (Antibacterials for systemic use), N03A (Antiepileptics), A10 (Drugs used in diabetes), G03 (Sex hormones and modulators of the genital system)	2004	2024	yes	China	no
60	2023	FAERS	Reporting Odds Ratio	Not used	Descriptive Statistics	N05CD08 (Midazolam)	Not informed	2023	yes	china	no
61	2022	FAERS	Bayesian Confidence Propagation Neural Network, Reporting Odds Ratio, Proportional Reporting Ratio, Multi-item Gamma Poisson Shrinker	Not used	Descriptive Statistics	S01LA04 (Ranibizumab), S01LA05 (Aflibercept), S01LA06 (Brolucizumab)	2004	2021	yes	china	no
62	2024	FAERS	Proportional Reporting Ratio, Reporting Odds Ratio, Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker	Not used	Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	L01FD04 (Trastuzumab deruxtecan), L01FX05(brentuximab vedotin), L01FX13 (Enfortumab vedotin), L01FX15 (Belantamab mafodotin), L01FX17 (Sacituzumab govitecan), L01FX26 (Mirvetuximab soravtansine)	2011	2023	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
63	2025	FAERS	Information Component, Reporting Odds Ratio, Bayesian Confidence Propagation Neural Network	Not used	Time-to-Onset (TTO) / Time-to-Event Analysis, Weibull Shape Parameter (WSP) Test, Wilcoxon Rank Sum Test, Kaplan-Meier method	L01EX18 (Avapritinib)	2020	2023	yes	China	no
64	2021	FAERS	Information Component, Reporting Odds Ratio	Logistic Regression	Not used	C09AA02 (Enalapril), C09AA03 (Lisinopril), C09AA04 (Perindopril), C09AA05 (Ramipril), C09AA06 (Captopril), C09AA07 (Benazepril), C09AA08 (Cilazapril), C09AA09 (Fosinopril), C09AA10 (Trandolapril), C09AA12 (Quinapril), C09AA13 (Moexipril), C09CA04 (Irbesartan)	2004	2020	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
65	2025	KIDS-KD/ KAERS	Information Component, Proportional Reporting Ratio, Reporting Odds Ratio, Bayesian Confidence Propagation Neural Network	Logistic Regression	Descriptive Statistics	J01C (Penicillin), "J01D (Cephalosporins, Monobactam, Carbapenem)", J01XA (Glycopeptide), "J01F (Macrolide, Lincomycin)", J01AA (Tetracycline), J01XX (Oxazolidinone), J01G (Aminoglycoside), J01M (Fluoroquinolone), J01XD (Nitroimidazole), J01EE (Sulfamethoxazole/Trimethoprim), D06AX09/R01AX06 (Mupirocin), J01XB02 (Polymyxin B), D06AX01 (Fusidic acid)	2013	2022	no	Korea	no
66	2022	VigiBase	Information Component	Logistic Regression	Descriptive Statistics	N02AX02 (Tramadol)	2014	2021	no	Iraq	no
67	2024	KIDS-KD/ KAERS	Information Component, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Chi-Square Test, Descriptive Statistics	A03FA01 (Metoclopramide), N05AH03 (Olanzapine), N05AH04(quetiapine), N05AX08 (Risperidone), N05AX12 (Aripiprazole)	2013	2022	no	Korea	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
68	2024	FAERS	MHRA Method, Reporting Odds Ratio	Not used	Not used	N05AE01 (Ziprasidone), N05AE05 (Lurasidone), N05AH02 (Clozapine), N05AH03 (Olanzapine), N05AH04(quetiapine), N05AH05 (Asenapine), N05AX08 (Risperidone), N05AX12 (Aripiprazole), N05AX13 (Paliperidone Palmitate), N05AX15 (Cariprazine), N05AX16 (Brexpiprazole)	2016	2022	yes	China	no
69	2023	FAERS	Information Component, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Not used	A02BC02 (Pantoprazole)	2006	2021	yes	India	no
70	2022	FAERS	Bayesian Confidence Propagation Neural Network, Reporting Odds Ratio, Proportional Reporting Ratio, Multi-item Gamma Poisson Shrinker, Information Component	Not used	Chi-Square Test, Fisher's Exact Test, Mann-Whitney U Test, Survival Analysis / Mortality Rate, Time-to-Onset (TTO) / Time-to-Event Analysis, Descriptive Statistics	J01XA01 (Vancomycin), J01XX08 (Linezolid)	2004	2021	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
71	2018	JADER	Not used	Association Rule Mining (ARM) – Apriori	Additive Interaction Model (AI), Combination Risk Ratio (CR), Multiplicative Interaction Model (MI)	M01AE14 (Loxoprofen)	2004	2015	no	Japan	no
72	2020	JADER	Chi-square Statistics Model, Shrinkage Measure Model, Proportional Reporting Ratio	Not used	Additive Interaction Model (AI), Cohen Kappa Coefficient, Combination Risk Ratio (CR), Multiplicative Interaction Model (MI)	A02BA01 (Cimetidine), A07AA10 (Colestimide), A16AA03 (L-glutamine), B03BA05 (Methylcobalamin), H03AA01 (Levothyroxine sodium hydrate), M01AB05 (Diclofenac sodium), M01CB03 (Auranofin), M04AA01 (Allopurinol), R05DB05 (Pentoxyverine citrate), S01BC08 (Pranoprofen), S01GX10 (Epinastine hydrochloride), S01XA01 (Sodium azulene sulfonate hydrate)	2004	2015	yes	Japan	no
73	2024	FAERS	Bayesian Confidence Propagation Neural Network, Proportional Reporting Ratio, Reporting Odds Ratio, Information Component	Not used	Descriptive Statistics	B01AC24 (Ticagrelor)	2010	2023	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
74	2025	FAERS	Empirical Bayes Geometric Mean, Information Component, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Logistic Regression, Time-to-Onset (TTO) / Time-to-Event Analysis, Weibull Shape Parameter (WSP) Test	L01BC02 (Fluorouracil), L01BC05 (Gemcitabine), L01BC06 (Capecitabine), L01CE02 (Irinotecan), L01FE01 (Cetuximab), L01FE02 (Panitumumab), L01FG01(Bevacizumab), L01XA03 (Oxaliplatin), V03AF03 (Leucovorin)	2019	2023	yes	China	no
75	2024	OpenVigil	Reporting Odds Ratio, Proportional Reporting Ratio	Not used	Descriptive Statistics, Chi-Square Test	B01AC23 (Cilostazol)	2004	2024	yes	China	no
76	2025	FAERS	Bayesian Confidence Propagation Neural Network, Reporting Odds Ratio, Proportional Reporting Ratio	Not used	Descriptive Statistics	L01EF01 (Palbociclib), L01EF02 (Ribociclib), L01EF03 (Abemaciclib)	2015	2023	yes	China	no
77	2025	FAERS	Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Chi-Square Test, Descriptive Statistics	H02CA02 (Osilodrostat)	Not informed	2024	yes	Romania	no
78	2024	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	ANOVA, Spearman Correlation Coefficient, Fisher’s Exact Test	L01EX17 (Capmatinib)	2020	2022	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
79	2018	FAERS	Proportional Reporting Ratio	Concomitant Medication Analysis, Molecular Data Integration, Receptor Agonism/Antagonism Cohort Analysis	Descriptive Statistics	N05AH04(quetiapine), N05AX08 (Risperidone), N05AH03 (Olanzapine), N05AX12 (Aripiprazole), N05AE01 (Ziprasidone) , N05AH02 (Clozapine), N05AE05 (Lurasidone), N05AH05 (Asenapine), N05AX13 (Paliperidone Palmitate)	2004	2016	yes	United States of America	no
80	2024	FAERS	Reporting Odds Ratio	Not used	Descriptive Statistics	J05AB16 (Remdesivir), L04AC07 (Tocilizumab)	2020	2022	yes	Brazil	no
81	2023	FAERS	Empirical Bayes Geometric Mean, Information Component, Proportional Reporting Ratio, Reporting Odds Ratio, Multi-item Gamma Poisson Shrinker	Not used	Chi-Square Test, Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	L01EX18 (Avapritinib)	2004	2022	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
82	2018	Netherlands Pharmacovigilance Centre Lareb Database	Reporting Odds Ratio	Bootstrap Resampling, Logistic Regression, Prediction Model-based Algorithm	Area Under the ROC Curve (AUC), Calibration Curve, Generalized Variance Inflation Factor (VIF), Hosmer-Lemeshow Goodness of Fit Test, Naranjo Score, Summary of Product Characteristics (SPC) as Gold Standard, Descriptive Statistics	000000 All drugs	Database inception	2016	no	Netherlands	no
83	2018	FAERS	Proportional Reporting Ratio	Decision Tree Analysis, Genetic Algorithm (GA), Multilabel Classification, Target Adverse-Event (TAE) Profile Analysis	F-measure maximization, Precision, Recall	A10BJ02 (Liraglutide), B01AX07 (Rivaroxaban), J05AG04 (Etravirine), L01EX02 (Pazopanib), L04AB05 (Certolizumab Pegol), N06AX16 (Desvenlafaxine)	2004	2015	yes	United States of America	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
84	2025	FAERS	Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Chi-Square Test, Descriptive Statistics	L01FF01 (Nivolumab), L01FF02 (Pembrolizumab), L01FF03 (Durvalumab), L01FF04 (Avelumab), L01FF04(Cemiplimab), L01FF05 (Atezolizumab), L01FX04 (Ipilimumab)	2003	2023	yes	China	no
85	2021	FAERS	Empirical Bayes Geometric Mean, MHRA Method, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	"C02C (ANTIADRENERGIC AGENTS, PERIPHERALLY ACTING)", "C09A (ACE INHIBITORS, PLAIN)", "C09CA (Angiotensin II receptor blockers (ARBs), plain)", C08CA (Dihydropyridine derivatives), "C07AB (Beta blocking agents, selective)", "C07AA (Beta blocking agents, non-selective)", "C03CA (Sulfonamides, plain)", "C03A (LOW-CEILING DIURETICS, THIAZIDES)", C03DA (Aldosterone antagonists)	2016	2019	yes	China	no
86	2023	FAERS	Reporting Odds Ratio	Not used	Crude Odds Ratio (OR) Calculation	J01XX08 (Linezolid)	2013	2021	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
87	2024	FAERS	Reporting Odds Ratio	KEGG Pathway Enrichment Analysis, Network Analysis / Hierarchical Clustering Analysis, Protein-Protein Interaction (PPI) Analysis	Not used	L01EF01 (Palbociclib), L01EF02 (Ribociclib), L01EF03 (Abemaciclib)	2014	2023	yes	China	no
88	2024	FAERS, VigiBase	Reporting Odds Ratio, Likelihood Ratio Test	Logistic Regression	Chi-Square Test, Kruskal-Wallis Test, Mann-Whitney U Test, Time-to-Onset (TTO) / Time-to-Event Analysis, Descriptive Statistics	L01XE24 (CAR T-cell therapy class)	2017	2023	yes	China	no
89	2023	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Chi-Square Test, Descriptive Statistics, Mann-Whitney U Test	H05AA03 (Abaloparatide)	2017	2022	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
90	2024	FAERS	Empirical Bayes Geometric Mean, Information Component, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Chi-Square Test, Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	R07AX02 (Ivacaftor), R07AX03 (Tezacaftor), R07AX04 (Elexacaftor)	2019	2022	yes	United States of America	no
91	2024	FAERS	MHRA Method, Reporting Odds Ratio	Not used	Descriptive Statistics	L01FF01 (Nivolumab), L01FF02 (Pembrolizumab), L01FF03 (Durvalumab), L01FF04 (Avelumab), L01FF04(Cemiplimab), L01FF05 (Atezolizumab)	2019	2023	yes	China	no
92	2019	FAERS, JADER	Reporting Odds Ratio	Association Rule Mining (ARM) – Apriori, Logistic Regression	Area Under the ROC Curve (AUC), Time-to-Onset (TTO) / Time-to-Event Analysis, Weibull Shape Parameter (WSP) Test	L01AX02 (Ifosfamide)	2004	2018	no	Japan	no
93	2022	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	L01FK01 (Olaparib)	2015	2021	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
94	2023	FAERS	Reporting Odds Ratio	Not used	Chi-Square Test, Descriptive Statistics, Mann-Whitney U Test, Time-to-Onset (TTO) / Time-to-Event Analysis, Weibull Shape Parameter (WSP) Test, Fisher's Exact Test	N05AH04(quetiapine)	2018	2022	yes	China	no
95	2019	FAERS	Proportional Reporting Ratio	Not used	Fisher's Exact Test	V10XX (Various therapeutic radiopharmaceuticals)	Not informed	2017	yes	Greece	no
96	2025	FAERS	Bayesian Confidence Propagation Neural Network, Proportional Reporting Ratio, Reporting Odds Ratio, Information Component	Not used	Chi-Square Test, Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	N02CD (Calcitonin gene-related peptide (CGRP) antagonists)	2020	2024	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
97	2025	FAERS	Empirical Bayes Geometric Mean, Information Component, Reporting Odds Ratio, Proportional Reporting Ratio	Not used	Kolmogorov–Smirnov Test (, Weibull Shape Parameter (WSP) Test, Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	L04AB01 (etanercept), L04AB02(infliximab), L04AB04 (adalimumab), L04AB05 (Certolizumab Pegol), L04AB06 (Golimumab)	2004	2024	yes	China	no
98	2025	FAERS	Bayesian Confidence Propagation Neural Network, Proportional Reporting Ratio, Reporting Odds Ratio	Logistic Regression	Descriptive Statistics, Naranjo Score	A10BH (Dipeptidyl peptidase 4 (DPP-4) inhibitors)	2004	2024	yes	Bahrain	no
99	2023	FAERS	Reporting Odds Ratio	Not used	Descriptive Statistics	A10BK (SGLT-2 Inhibitors), A10BJ (Glucagon-like peptide-1 (GLP-1) analogues)	2012	2022	yes	United States of America	no
100	2023	OpenVigil	Reporting Odds Ratio, Proportional Reporting Ratio	Not used	Descriptive Statistics, Chi-Square Test	M05BA08 (Zoledronic acid), M05BX04(denosumab)	2004	2022	yes	China	no
101	2017	FAERS	Information Component, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	N06AX26 (Vortioxetine)	2015	2016	yes	India	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
102	2018	FAERS	Information Component, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	A08AA10 (Sibutramine), A10BG02 (Rosiglitazone), A10BH01 (Sitagliptin), A10BK02 (Canagliflozin), N02AC05 (Propoxyphene)	2006	2015	yes	India	no
103	2024	FAERS	Empirical Bayes Geometric Mean, Information Component, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	A02BC (Proton pump inhibitors)	2013	2022	yes	China	no
104	2024	FAERS	Case/non-case analysis, Reporting Odds Ratio, Information Component	Logistic Regression	Descriptive Statistics	J05AE (Protease inhibitors)	2023	2023	yes	China	no
105	2024	FAERS	Bayesian Confidence Propagation Neural Network, Reporting Odds Ratio, Information Component	Not used	Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	L01XM (Isocitrate dehydrogenase (IDH) inhibitors)	2017	2024	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
106	2025	FAERS	Bayesian Confidence Propagation Neural Network, Reporting Odds Ratio, Information Component	Not used	Descriptive Statistics	C03DA01 (Spironolactone), G03BA03 (Testosterone), G04CA02 (Tamsulosin), G04CB01 (Finasteride), G04CB02 (Dutasteride), L01CA01 (Vinblastine), L04AA06 (Mycophenolate mofetil), L04AA27 (Fingolimod), M04AA01 (Allopurinol), N03AF01 (carbamazepine), N03AF02 (Oxcarbazepine), N03AG01 (valproic acid), N03AX14 (Levetiracetam), N05AE01 (Ziprasidone), N05AH02 (Clozapine), N05AH03 (Olanzapine), N05AX08 (Risperidone), N05AX12 (Aripiprazole), N05AX13 (Paliperidone Palmitate), N06BA09 (Atomoxetine)	2004	2023	yes	China	no
107	2025	FAERS	Bayesian Confidence Propagation Neural Network, Empirical Bayes Geometric Mean, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Not used	"V08AB (Watersoluble, nephrotropic, low osmolar X-ray contrast media)"	2004	2023	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
108	2025	FAERS	Reporting Odds Ratio	Not used	Descriptive Statistics, Kaplan-Meier method, Time-to-Onset (TTO) / Time-to-Event Analysis	L01FF01 (Nivolumab), L01FF02 (Pembrolizumab), L01FF03 (Durvalumab), L01FF04 (Avelumab), L01FF04(Cemiplimab), L01FF05 (Atezolizumab), L01FF07 (Dostarlimab), L01FF13 (Toripalimab)	2011	2024	yes	China	no
109	2025	FAERS, JADER	Reporting Odds Ratio	Not used	Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	L01FX05 (Brentuximab vedotin)	2004	2023	no	China	no
110	2022	FAERS	Reporting Odds Ratio, Case/non-case analysis	Not used	Not used	J02AC04(voriconazole)	2004	2021	yes	Japan	no
111	2024	FAERS	Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Chi-Square Test, Descriptive Statistics	L01FB01 (Inotuzumab ozogamicin), L01FD03 (Trastuzumab emtansine), L01FD04 (Trastuzumab deruxtecan), L01FX02 (Gemtuzumab ozogamicin), L01FX05 (Brentuximab vedotin), L01FX13 (Enfortumab vedotin), L01FX14 (Polatuzumab vedotin)	2004	2022	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
112	2021	FAERS	Bayesian Confidence Propagation Neural Network, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Not used	J05AE (Protease inhibitors)	2006	2019	yes	China	no
113	2025	OpenVigil	Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Not used	J02AC (Triazole and tetrazole derivatives)	2002	2023	yes	China	no
114	2024	JADER	Reporting Odds Ratio	Logistic Regression	Chi-Square Test, Descriptive Statistics, Hosmer-Lemeshow Goodness of Fit Test	Glycyrrhiza	2004	2022	yes	Japan	no
115	2025	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	L04AC12 (Brodalumab)	2017	2023	yes	China	no
116	2024	FAERS	Bayesian Confidence Propagation Neural Network, Empirical Bayes Geometric Mean, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	C09DX04 (Valsartan and Sacubitril)	2015	2023	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
117	2018	FAERS, Taiwan's National Health Insurance Research Database (NHIRD)	Case-Only Odds Ratio (OR) Analysis	Not used	Not used	A10 (Drugs used in diabetes)	2004	2014	no	China	no
118	2023	FAERS	Case-Only Odds Ratio (OR) Analysis, Case/non-case analysis	Not used	Chi-Square Test, Descriptive Statistics, Kruskal-Wallis Test, Mann-Whitney U Test, Time-to-Onset (TTO) / Time-to-Event Analysis	C09 (Agents acting on the renin-angiotensin system)	2004	2020	yes	China	no
119	2024	FAERS	Reporting Odds Ratio, Proportional Reporting Ratio, Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker	Not used	Chi-Square Test, Weibull Shape Parameter (WSP) Test	L01EG08 (Tivozanib)	2021	2023	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techiniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
120	2024	FAERS	Bayesian Confidence Propagation Neural Network, Proportional Reporting Ratio, Reporting Odds Ratio, Information Component	Not used	Descriptive Statistics	M05B (Drugs affecting bone structure and mineralization)	2003	2023	yes	China	no
121	2024	FAERS	MHRA Method, Reporting Odds Ratio, Proportional Reporting Ratio	Not used	Chi-Square Test, Descriptive Statistics	"J05AR18 (emtricitabine, tenofovir alafenamide, elvitegravir and cobicistat)"	2015	2023	yes	China	no
122	2023	FAERS	Bayesian Confidence Propagation Neural Network, MHRA Method, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics, Chi-Square Test	L01FX06 (Dinutuximab/Dinutuximab β), L01FX21 (Naxitamab)	2015	2023	yes	China	no
123	2022	FAERS	Information Component, Reporting Odds Ratio	Not used	Descriptive Statistics	J01AA02 (Doxycycline), J01AA08 (Minocycline), J01AA12 (Tigecycline)	2004	2020	yes	China	no
124	2023	FAERS	Information Component, Reporting Odds Ratio	Statistical Shrinkage Transformation	Descriptive Statistics	B02BX06 (Emicizumab)	2017	2021	yes	China	no
125	2021	FAERS	Information Component, Proportional Reporting Ratio, Case-Only Odds Ratio (OR) Analysis	Not used	Chi-Square Test, Time-to-Onset (TTO) / Time-to-Event Analysis	A02BC (Proton pump inhibitors)	2004	2020	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
126	2021	FAERS	Reporting Odds Ratio	Not used	Descriptive Statistics	A02BC (Proton pump inhibitors)	2004	2019	yes	China	no
127	2023	FAERS	Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	L04AF03 (Upadacitinib)	2004	2023	yes	China	no
128	2022	FAERS	Information Component, Reporting Odds Ratio	Not used	Chi-Square Test, Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	L01F (Monoclonal antibodies and antibody drug conjugates)	2004	2021	yes	China	no
129	2024	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	A10BJ06 (Semaglutide)	2019	2023	yes	China	no
130	2021	FAERS	Empirical Bayes Geometric Mean, Proportional Reporting Ratio, Reporting Odds Ratio, Information Component	Not used	Descriptive Statistics	N03AF01 (carbamazepine), N03AF02 (Oxcarbazepine)	2004	2019	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
131	2025	FAERS	Bayesian Confidence Propagation Neural Network, Empirical Bayes Geometric Mean, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	L04AG05 (Vedolizumab)	2014	2023	yes	China	no
132	2023	FAERS	Information Component, Reporting Odds Ratio	Not used	Descriptive Statistics	N05A (Antipsychotics)	2004	2021	yes	China	no
133	2022	FAERS	Information Component, Reporting Odds Ratio	Not used	Descriptive Statistics	L01FX13 (Enfortumab vedotin)	2019	2021	yes	China	no
134	2025	FAERS	Empirical Bayes Geometric Mean, Information Component, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	L04AG10 (Efgartigimod alfa)	2022	2024	yes	China	no
135	2024	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	J01DD52 (Ceftazidime and beta-lactamase inhibitor)	2015	2022	yes	China	no
136	2024	FAERS	Proportional Reporting Ratio, Reporting Odds Ratio	Logistic Regression	Descriptive Statistics	L02BA03 (Fulvestrant)	2004	2023	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
137	2025	FAERS	Proportional Reporting Ratio, Reporting Odds Ratio, Shrinkage Measure Model	Not used	Not used	J01XX08 (Linezolid)	2014	2024	yes	China	no
138	2023	FAERS	Empirical Bayes Geometric Mean, Information Component, Reporting Odds Ratio, Bayesian Confidence Propagation Neural Network	Not used	Descriptive Statistics	L01 (Antineoplastic agents)	2004	2021	yes	China	no
139	2024	FAERS	Bayesian Confidence Propagation Neural Network, Proportional Reporting Ratio, Reporting Odds Ratio, Empirical Bayes Geometric Mean, Information Component	Not used	Time-to-Onset (TTO) / Time-to-Event Analysis, Descriptive Statistics, Chi-Square Test	L01EX08(Lenvatinib)	2015	2023	yes	China	no
140	2025	OpenVigil	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio, Information Component	Not used	Not used	L04AA32 (Apremilast)	2014	2023	yes	China	no
141	2018	FAERS	Reporting Odds Ratio	Not used	Chi-Square Test	J05AB01 (aciclovir), J05AB11 (valaciclovir), M01A (AINEs)	2004	2012	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
142	2023	FAERS	sROR (compressed reporting odds ratio)	Not used	Not used	L01FG01(Bevacizumab), S01LA04 (Ranibizumab), S01LA05 (Aflibercept)	2017	2021	yes	China	no
143	2020	FAERS	Information Component, Proportional Reporting Ratio, Reporting Odds Ratio, Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker	Not used	Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis, Weibull Shape Parameter (WSP) Test	L01XK03 (Rucaparib)	2017	2022	yes	China	no
144	2024	FAERS	Information Component, Reporting Odds Ratio	Not used	Descriptive Statistics	C10AA (HMG CoA reductase inhibitors)	2004	2022	yes	China	no
145	2024	FAERS	Reporting Odds Ratio	Not used	Descriptive Statistics	L04AG13 (Teprotumumab)	2020	2023	yes	China	no
146	2024	FAERS	Empirical Bayes Geometric Mean, Information Component, Reporting Odds Ratio	Not used	Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	N07XX13 (Valbenazine	2017	2023	yes	China	no
147	2025	FAERS	Bayesian Confidence Propagation Neural Network, Gamma Poisson Shrinker (GPS), Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	"J05AR18 (emtricitabine, tenofovir alafenamide, elvitegravir and cobicistat)"	2016	2023	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
148	2025	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	J01FA10 (Azithromycin)	2004	2024	yes	China	no
149	2021	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio, Information Component	Not used	Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	A10BK (SGLT-2 Inhibitors)	2004	2019	yes	China	no
150	2022	FAERS	Bayesian Confidence Propagation Neural Network, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	L01FK01 (Olaparib), L01XK02(Niraparib), L01XK03 (Rucaparib), L01XK04(Talazoparib)	2014	2021	yes	China	no
151	2024	FAERS	Bayesian Confidence Propagation Neural Network, Empirical Bayes Geometric Mean, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics	L04AH02(everolimus)	2009	2022	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
152	2024	OpenVigil	Bayesian Confidence Propagation Neural Network, Reporting Odds Ratio	Not used	Not used	N06AB03 (fluoxetine), N06AB04(citalopram), N06AB05 (paroxetine), N06AB06 (Sertraline), N06AB08(Fluvoxamine),N06AB10(Escitalopram)		2023	yes	China	no
153	2025	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio	Not used	Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	M03BX01 (Baclofen)	2004	2024	yes	China	no
154	2022	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio, Information Component	Not used	Descriptive Statistics, Time-to-Onset (TTO) / Time-to-Event Analysis	J02A (Antimycotics for systemic use)	2004	2021	yes	China	no
155	2024	FAERS	Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Proportional Reporting Ratio, Reporting Odds Ratio, Information Component	Not used	Descriptive Statistics	N05AX15 (Cariprazine)	2015	2023	yes	China	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
156	2018	FAERS, JADER	sROR (compressed reporting odds ratio), Information Component	Not used	Descriptive Statistics, Chi-Square Test, Fisher's Exact Test, Mann-Whitney U Test	L02AE02 (leuprorelin), L02AE04 (Triptorelin), L02AE03 (Goserelin), L02AE05 (Histrelin), L02AE01 (Buserelin)	2004	2023	yes	China	no
157	2018	FAERS	Reporting Odds Ratio	Not used	Descriptive Statistics	000000 All drugs	2004	2016	yes	Italy	no
158	2022	FAERS	Reporting Odds Ratio	Not used	Time-to-Onset (TTO) / Time-to-Event Analysis, Mann-Whitney U Test, Kruskal-Wallis Test, Chi-Square Test, Fisher's Exact Test	A10BK (SGLT-2 Inhibitors)	2004	2020	yes	China	no
159	2021	KIDS-KD/ KAERS	Reporting Odds Ratio, Proportional Reporting Ratio, Information Component, Bayesian Confidence Propagation Neural Network	Not used	Not used	J01AA02 (Doxycycline)	2014	2018	no	Korea	no

article_id	year_of_publication	primary_data_source	disproportionality_techniques	data_mining_algorithms_techiniques	statistical_or_other_methods	therapeutic_class_or_drug	start_data_collection	end_data_collection	open_access_data	country	post_processed_dataset_available
160	2021	FAERS	Reporting Odds Ratio, Proportional Reporting Ratio, Bayesian Confidence Propagation Neural Network, Multi-item Gamma Poisson Shrinker, Information Component	Not used	Time-to-Onset (TTO) / Time-to-Event Analysis	L01EK01 (Axitinib)	2018	2021	yes	China	no
161	2020	FAERS	Case/non-case analysis, Reporting Odds Ratio	Not used	Descriptive Statistics	L04AF02 (Baricitinib	2018	2020	yes	China	no
162	2025	FAERS	Not used	K-means, Hierarchical K-means (HK-means), Spectral Clustering, Gaussian Mixture Model (GMM), Isolation Forest (iF), Extended Isolation Forest (eiF)	Calinski–Harabasz Index, Davies–Bouldin Index, Dunn Index, Baker-Hubert Gamma Index	N03A (Antiepileptics), J01 (Antibacterials for systemic use)	2013	2022	yes	United States of America	no

S3: Code Repository

All scripts and materials used in this scoping review are publicly available in a GitHub repository.

Repository: https://github.com/carolinajacoby/pharmacovigilance-scoping-review

Technical information:

Programming language: Python 3.10

Main libraries: pandas, numpy, matplotlib, seaborn

Development environment: Google Colab

Version control: Git/GitHub

Detailed instructions for installation, dependencies, and execution guidelines are available in the repository's README file. This material aims to ensure the reproducibility of the computational analyses, in accordance with PRISMA-ScR and best practices in open science.

Repository Structure

Directory	**Description**
data	Contains the data used in the review.
data/excluded	CSV files with excluded articles and reasons for exclusion.
data/processed	Processed data with information from included articles.
data/supplementary_material	Supplementary materials from the review in PDF format.
docs	Contains the scoping review protocol in PDF format.
prism	Contains the PRISMA diagram in PDF format.
scripts	Contains the code or notebooks used for data analysis.
LICENSE	File with the repository license
README	Project description, usage instructions, and dependencies.